Development of a Knowledge Graph Front-End for the Islamic Scholarly Network

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***Abstract*—The hadith *sanad* is an Islamic scholarly network formed by chains of teacher–student relationships among narrators across generations. Its relational structure lends itself naturally to directed-graph modeling, enabling computational measurement of each narrator's role and influence. Prior studies have applied Knowledge Graph (KG) and Social Network Analysis (SNA) methods to *sanad* networks, yet their outputs are largely static and accessible only through specialized desktop software, keeping the results out of reach for students, researchers, and the general public who must install particular tools and possess the technical knowledge to operate them. This paper presents the development of a publicly accessible, web-based front-end that integrates an interactive KG of the hadith narrator network with five SNA centrality metrics. From the Sanadset 650K corpus (650,986 hadith entries from 926 classical Arabic books), 487,451 cleaned *sanad* chains were modeled as a weighted directed graph of 177,047 narrator nodes and 776,798 narration edges. In-degree, out-degree, and eigenvector centrality were computed on the full graph, while closeness and betweenness centrality were computed on a 14,695-node strongest-weight subgraph. The analysis identified Abu Hurairah as the most central narrator in four of five metrics, while Sufyan ranked highest in out-degree. The results were stored in a Supabase (PostgreSQL) backend and served through a React dashboard with four pages: Narrator Relation Graph, Narrator Statistics, Relation Table, and Centrality Table; the interactive graph is rendered on an HTML5 Canvas with a custom force-directed layout. Black-box testing across 14 scenarios achieved a 100% success rate. The system fills the gap left by prior work by turning a large-scale *sanad* KG into an interactive front-end that is freely explorable without any software installation.**

***Index Terms*—knowledge graph, hadith sanad, social network analysis, narrator centrality, network visualization, interactive dashboard**

I. Introduction

Hadith, the recorded sayings, actions, and tacit approvals of the Prophet Muhammad, is the second source of Islamic law after the Qur'an [1], [2]. Structurally, a hadith consists of two components: the *matn* (the text itself) and the *sanad*, the continuous chain of narrators who transmitted the text across generations. The *sanad* forms an Islamic scholarly network built from teacher–student relationships, in which each narrator is a link in the intergenerational transfer of religious knowledge [3]. Because of this relational structure, a *sanad* is naturally represented as a directed graph, with narrators as nodes and teacher–student relations as edges [4].

Not all narrators play an equal role. Some occupy strategic positions by receiving hadith from many teachers and transmitting them to many students, becoming convergence points traversed by diverse *sanad* paths. Such narrators, known as *common links*, underscore the importance of identifying the most influential and most connective narrators [5]. Classical scholarship performs this identification manually through *isnad*-bundle analysis, but when the scope spans hundreds of thousands of narrators with highly complex relations, manual analysis becomes inadequate, motivating quantitative and computational approaches that measure each narrator's position and influence at scale [4].

This need has been addressed over the past decade by studies that analyze the narrator network quantitatively. Several works model historical social networks of narrators and apply network analysis to reconstruct temporal information from the transmission structure [6], while others represent the narrator network as a Knowledge Graph (KG) and analyze it with network software [4]. Centrality metrics such as in-degree, out-degree, betweenness, and closeness have been used to identify the most influential narrators [7]. These studies confirm that the narrator network can be modeled and analyzed computationally at scale. Their outputs, however, are generally static and depend on specialized desktop software such as Gephi and Cytoscape, requiring installation and technical expertise that restrict access to a narrow audience of specialists [4]. Students, researchers, and the general public who lack these tools and skills therefore cannot explore the Islamic scholarly network and its centrality insights interactively.

The KG representation used in those studies nonetheless offers a foundation that can be developed into a more interactive and widely accessible system. A KG captures entities and the relations among them as a graph, presenting complex, connected data in a semantic and structured manner; it has been widely adopted across domains such as education and tourism [8]. For hadith studies, this representation is well suited to modeling the narration network, with narrators as entities and teacher–student relations as the connecting edges [4]. To turn such a network into knowledge that can be explored easily, an interactive web-based front-end is required for three reasons: interactivity (users need to zoom, filter, search, and traverse nodes directly rather than view a static image), accessibility (a web interface requires no software installation and reaches a wide audience), and presentation (rendering an enormous network all at once harms readability, so centrality scores can be used to emphasize the most central nodes [9]).

This paper presents the development of a web-based front-end that integrates an interactive KG of the hadith narrator network with SNA centrality analysis, built on the publicly available Sanadset 650K dataset [10]. The contributions are threefold: (i) construction of a large-scale weighted directed KG of the *sanad* network through systematic preprocessing and directed-graph modeling of teacher–student relations; (ii) quantitative identification of each narrator's role and influence using five SNA centrality metrics—in-degree, out-degree, eigenvector, closeness, and betweenness; and (iii) an interactive, publicly deployed web dashboard that makes the narrator KG and its centrality analysis freely explorable without any software installation, addressing the static and desktop-bound limitations of prior work.

II. Related Work

Computational research on the hadith *sanad* has progressed from dataset construction toward graph-based modeling and SNA. Mghari *et al.* built Sanadset 650K, a corpus of 650,986 entries from 926 classical Arabic books, in which each entry separates the full hadith text, *matn*, narrator list, and *sanad* ordering [10]. Its scale and coverage make it the most representative source for *sanad* transmission research, though it is provided as raw data without graph visualization or network analysis. The same group later proposed Narrator2Vec, a Skip-Gram Word2Vec adaptation that learns vector representations of narrator names and predicts missing narrators in broken chains [7]; it does not, however, produce per-narrator quantitative scores that users can compare.

Farooqi *et al.* introduced Multi-IsnadSet, representing the Sahih Muslim transmission network as a directed graph of 2,092 nodes and 77,797 edges, built and analyzed with NetworkX, Neo4j, Gephi, and Cytoscape [4]. They demonstrated that a *sanad* network can be modeled as a KG and analyzed with degree, betweenness, and eigenvector centrality, but the resulting visualizations are static and depend on desktop software. Shuaib *et al.* applied trophic analysis to a 49,309-narrator hadith network and showed that the network is hierarchical and sequential, recovering temporal information from structure [6], yet without centrality metrics or an interactive interface. Mahmoud *et al.* proposed NarratorsKG, the first KG dedicated to hadith narrators (79,256 nodes, 197,183 edges), used for narrator disambiguation with graph-query embeddings and AraBERT [11]; it is an internal NLP resource rather than an end-user system. Hendawi *et al.* used machine learning to verify *sanad* continuity from narrator biographical data [12], reinforcing the value of structured narrator data.

A parallel line of work visualizes narrator chains as graphs. Saraçoğlu converted 2,258 *isnad* chains into source–target pairs and visualized them with Pajek, Gephi, and Isnalyser [13]. HadithNet automated narrator-name identification and chain generation on a web system using Django, MySQL, and Graphviz, reaching 99.37% name precision but struggling with complex branching chains [14]. Ahmad *et al.* showed that transforming *isnad* into directed graphs enables centrality, clustering, and bridge-narrator analysis [15], and the KASHAF system offered interactive Sankey-flow visualization over 60,000 hadith [16], though focused on retrieval and *takhreej* classification rather than narrator-centrality exploration.

Across these studies, two gaps persist. Dataset-oriented works yield large corpora without an exploration interface, while analytical works produce models operable only through desktop software or technical expertise. No prior study integrates a narrator KG with SNA centrality metrics in a single web interface accessible without specialized software. This work fills that gap.

III. Materials and Methods

The research was conducted in four stages: preprocessing of the Sanadset 650K dataset, KG construction from teacher–student relations, network analysis using five centrality metrics, and development of an interactive web dashboard. The analytical pipeline was implemented in Python 3.13 with NetworkX 3.6, and the front-end in React 18 with a Supabase (PostgreSQL) backend.

A. Dataset

The primary input is `sanadset.csv` from Sanadset 650K [10], a static CSV of 650,986 hadith records drawn from 926 classical Arabic books. Each record contains the source book, hadith number, hadith text, *matn*, narrator list, and *sanad* length. Only the *sanad* column is used, as it carries the narrator chain required for graph construction and centrality computation.

B. Data Preprocessing

Because narrator names in classical texts are written inconsistently, a single narrator may otherwise be split into several nodes, distorting the graph and its centrality values. Seven preprocessing steps were applied:

1) **Removal of chainless rows.** Rows marked `No SANAD` carry no narration relation and were removed, eliminating 159,558 rows.
2) **Diacritic (*harakat*) removal.** Arabic vocalization marks were stripped via regular expressions so that the same name written with or without diacritics maps to one entity.
3) **Arabic-letter normalization.** Unicode variants of *alif*, *ya*/*alif maqsura*, and *ta marbuta* were unified to canonical forms.
4) **Kinship-term resolution.** Contextual terms meaning "his father," "his mother," or "his grandfather" were

TABLE I
DATA REDUCTION ACROSS PREPROCESSING STAGES

Stage	Process	Rows	Removed
1	Raw data	650,986	—
2	Remove chainless rows (No SANAD)	491,428	159,558
3	Remove single-*Abu* rows	488,195	3,233
4	Remove kinship-led rows	487,451	744
	Final	**487,451**	**163,535**

replaced by explicit constructions referencing the preceding narrator, preventing generic merge nodes.

5) **Transmission-term removal.** Narration formulas (e.g., *haddathana*, *akhbarana*, *'an*) were removed so they do not form spurious nodes.
6) ***Abu* variant normalization.** Grammatical forms *Abu*/*Aba*/*Abi* were unified to the nominative *Abu*.
7) **Non-Arabic character validation.** Characters outside the Arabic Unicode range were removed; none remained after the prior steps.

Additional removal of ambiguous rows (single *Abu* elements and rows beginning with kinship terms) left 487,451 clean *sanad* chains for graph construction, as summarized in Table I.

C. Knowledge Graph Construction

The *sanad* network is modeled as a KG of structured triples (Student, narrates-from, Teacher). Each chain of n narrators is decomposed into $n-1$ triples, one per adjacent pair. Because transmission flows directionally from student to teacher, the network is built as a directed graph using `nx.DiGraph()`, with each student node as the source and each teacher node as the target. Self-loops are removed.

Each edge carries two weight attributes. *Weight* is the frequency with which a teacher–student pair occurs across the corpus; a higher weight indicates a stronger, more frequently attested transmission. *Inverse Weight*, the reciprocal $1/\text{Weight}$, serves as the edge distance for path-based centrality: frequent relations imply closer narrators (shorter distance), while rare relations imply greater distance. The final graph comprises **177,047 narrator nodes** and **776,798 narration edges**. Narrator names in Arabic script were additionally transliterated to Latin script for readability.

D. Centrality Metrics

Five SNA centrality metrics [17] were computed for every node, each capturing a distinct narrator role.

In-degree centrality measures incoming edges, i.e., how many students narrate from a narrator (role as a source/teacher):

$$C_D^{in}(v) = \frac{\deg^{in}(v)}{n-1}. \tag{1}$$

Out-degree centrality measures outgoing edges, i.e., how many teachers a narrator received from (role as an active student):

$$C_D^{out}(v) = \frac{\deg^{out}(v)}{n-1}. \tag{2}$$

Eigenvector centrality measures influence by connection quality—being linked to other influential narrators:

$$C_E(v) = \frac{1}{\lambda} \sum_{u \in N(v)} A_{vu}\, C_E(u), \tag{3}$$

where λ is the largest eigenvalue of the adjacency matrix A and $N(v)$ is the neighbor set of v.

Closeness centrality measures proximity to all other nodes by shortest path, using *Inverse Weight* as distance:

$$C_C(v) = \frac{n-1}{\sum_{u \neq v} d(v,u)}. \tag{4}$$

Betweenness centrality measures how often a node lies on shortest paths between other pairs, marking bridge narrators:

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}, \tag{5}$$

where σ_{st} is the number of shortest s–t paths and $\sigma_{st}(v)$ the number passing through v.

In-degree, out-degree, and eigenvector centrality are computationally light and were computed on the full graph. Closeness and betweenness require all-pairs shortest-path traversal, which is intractable on the full graph; they were computed on a representative subgraph of the 50,000 strongest-weight edges (14,695 nodes), preserving the principal transmission backbone.

E. System Architecture and Dashboard

The system follows a three-tier architecture (Fig. 1) separating heavy computation from data presentation. The *data processing layer* runs offline in Python, performing preprocessing, KG construction, and centrality computation, then exporting two CSV files: `sentralitas.csv` (per-narrator centrality scores) and `edge_slim.csv` (weighted relations). The *data storage layer* is Supabase (PostgreSQL), holding a CENTRALITY table (nodes) and an EDGE table (relations) linked by narrator ID, queryable via REST API. The *presentation layer* is a read-only Single Page Application built with React, TypeScript, and Vite, styled with TailwindCSS, that retrieves data through `@supabase/supabase-js` and renders four pages. Statistical charts use Recharts, while the interactive network graph is rendered directly on an HTML5 Canvas with a custom force-directed layout. Placing all computation in the processing layer ensures centrality is computed once on the backend rather than on the client, keeping the dashboard lightweight and responsive.

The force-directed layout combines three forces per animation frame: a Coulomb-like *repulsion* between nearby nodes, a Hooke-like *spring* force along edges toward an ideal

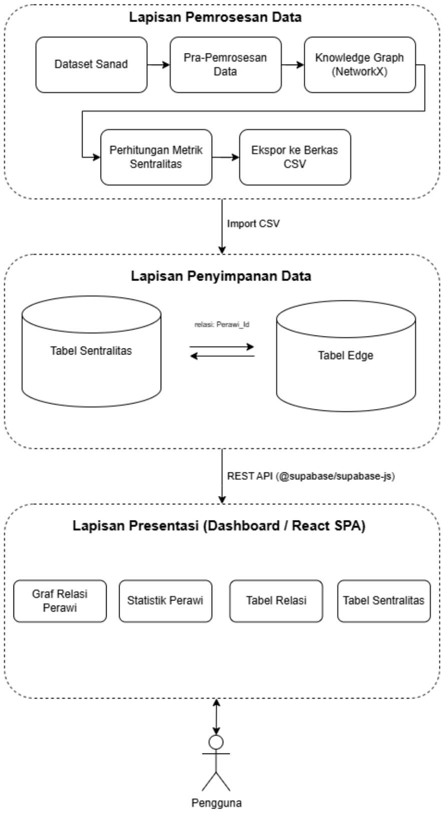

Fig. 1. Three-tier system architecture separating offline analysis, centralized storage, and the interactive presentation layer.

length, and a *radial constraint* that pulls each node toward a concentric ring keyed to its centrality rank, so the most central narrators settle near the graph center. Centrality is further encoded visually through node size and color (rank groups), and edges are drawn as arrows pointing from student to teacher. The dashboard was validated with black-box testing.

IV. Results and Discussion

A. Graph Construction

Normalization yielded 177,289 unique narrator names; of these, 177,047 participate in at least one narration relation and form the graph nodes, the remaining 242 being isolated names (single-narrator chains or names dropped during self-loop and empty-name cleaning). The resulting directed graph contains 177,047 nodes and 776,798 edges. The strongest relation is Ibnu Umar $\leftarrow$ Nafi' with a weight of 11,346, indicating that Nafi' narrated from Ibnu Umar that many times in the corpus; such high-weight pairs form a dominant transmission backbone that recurs across many chains.

B. Centrality Analysis

Table II lists the top narrators for each metric. Abu Hurairah ranks first in four of five metrics—in-degree, eigenvector, closeness, and betweenness—identifying him as the most

TABLE II
Top Narrators by Centrality Metric

Rank	Metric	Narrator	Score
1	In-degree	Abu Hurairah	0.0199
2	In-degree	Ibnu Abbas	0.0150
3	In-degree	Aisyah	0.0132
1	Out-degree	Sufyan	0.0135
2	Out-degree	Abu Abdullah Al-Hafizh	0.0127
3	Out-degree	Syu'bah	0.0124
1	Eigenvector	Abu Hurairah	0.5195
2	Eigenvector	Aisyah	0.4836
3	Eigenvector	Ibnu Umar	0.3821
1	Closeness	Abu Hurairah	1.0000
2	Closeness	Aisyah	0.9965
3	Closeness	Ibnu Abbas	0.9864
1	Betweenness	Abu Hurairah	0.0500
2	Betweenness	Syu'bah	0.0445
3	Betweenness	Sufyan	0.0384

central and most influential narrator in the network. This is consistent with his historical status as the most prolific companion narrator. Sufyan ranks first in out-degree, reflecting a distinct role as an exceptionally active recipient who narrated from many teachers; the divergence between the in-degree and out-degree rankings confirms that dominant sources of narration are not necessarily the most active receivers.

The eigenvector scores drop sharply after the top four narrators (from 0.2659 for Ibnu Abbas to 0.1723 for Anas), indicating that influence is concentrated in a small, tightly interconnected core. Closeness scores among the top ten are very high and closely spaced (0.9630–1.0000) because, within the strongest-weight subgraph, central narrators reach others through very short paths. In betweenness, the recurrence of Abu Hurairah, Syu'bah, and Sufyan among the top ranks signals a dual role as both influential and connective nodes. Together, the five metrics provide complementary, quantitatively distinguishable views of narrator roles.

C. Dashboard Implementation

The analytical results were stored in Supabase and served through the React dashboard. Fig. 2 shows the Narrator Relation Graph page, the core of the system, rendering narrators as nodes and teacher–student relations as directed edges on an HTML5 Canvas. A left filter panel provides search and the five centrality metrics; selecting a metric drives node size and color so that the most central narrators appear largest and centered. For readability the view displays a representative subset of the top 50 narrators with 1,000 relations. Selecting a node opens a right-hand detail panel with the narrator's connected teacher and student counts, all five centrality values, and the linked narrators, while highlighting the node's direct relations. The

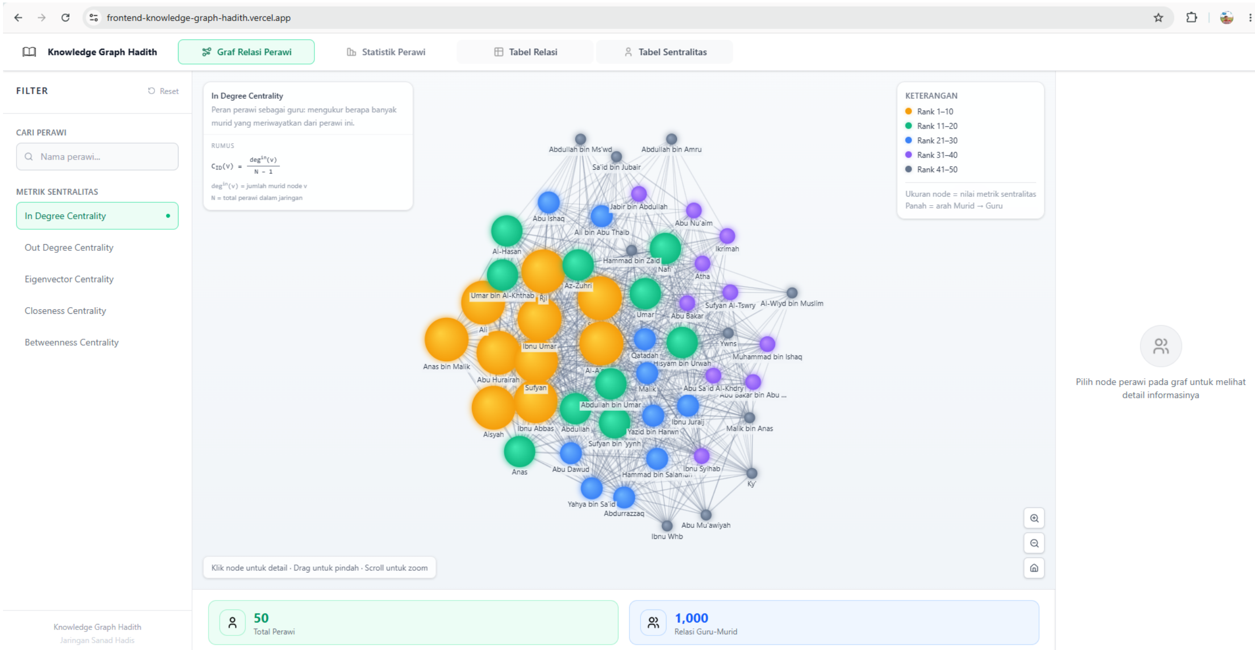

Fig. 2. Narrator Relation Graph page of the dashboard. Node size and color encode centrality rank (In-degree selected); arrows point from student to teacher. The left panel filters by metric and search; the right panel shows per-narrator detail when a node is selected.

graph supports zoom, pan, drag, node selection, and auto-fit reset.

The remaining three pages complement this view. The *Narrator Statistics* page summarizes the network with total-narrator (177,047) and total-relation (776,798) indicator cards and five bar charts of the top ten narrators per metric. The *Relation Table* page lists all 776,798 weighted teacher–student relations with pagination, sorting, and search; by default it is sorted by weight, placing Ibnu Umar–Nafi' (11,346) at the top. The *Centrality Table* page lists all 177,047 narrators with their five centrality scores, also paginated, sortable, and searchable. The dashboard was deployed online and is publicly accessible without installation.

D. System Testing

Black-box testing covered 14 scenarios spanning menu navigation, data loading, graph rendering, node selection, metric filtering, narrator search, zoom, drag and pan, auto-fit reset, the statistics page, the relation and centrality tables, pagination, and column sorting. All 14 scenarios produced the expected output, yielding a 100% success rate and confirming that the dashboard functions as designed.

E. Comparison with Prior Work

Table III positions this work against prior *sanad* network studies. Earlier work successfully models and analyzes narrator networks at scale, but the outputs are typically static and tied to desktop software such as Gephi and Cytoscape, restricting access to expert users [4], [6]. The principal advantage of this work is the presentation of the analysis as an interactive, publicly accessible web dashboard that integrates

TABLE III
COMPARISON WITH PRIOR SANAD NETWORK STUDIES

Study	KG	SNA	Web	Public
Farooqi *et al.* [4]	Yes	Partial	No	No
Shuaib *et al.* [6]	No	No	No	No
Mahmoud *et al.* [11]	Yes	No	No	No
Ahmad *et al.* [15]	Yes	Yes	No	No
KASHAF [16]	Yes	No	Yes	Partial
This work	**Yes**	**Yes**	**Yes**	**Yes**

a large-scale narrator KG with all five centrality metrics and exploration features (search, metric filtering, pagination, sorting, and teacher–student detail) in a single interface.

This work also has limitations. Closeness and betweenness centrality were computed on the strongest-weight subgraph rather than the full graph due to computational cost; the dataset is static and not automatically updated; and the analysis focuses on network structure without assessing *sanad* authenticity from the perspective of hadith science.

V. Conclusion

This paper presented an interactive, web-based KG front-end for the hadith narrator network with integrated SNA. From the Sanadset 650K corpus, 487,451 cleaned *sanad* chains were modeled as a weighted directed graph of 177,047 narrator nodes and 776,798 narration edges. Five centrality metrics distinguished narrator roles, identifying Abu Hurairah as the most central narrator in four of five metrics and Sufyan as the most active recipient by out-degree. The results were served through a React dashboard backed by Supabase, with an interactive force-directed graph rendered on an HTML5 Canvas and four exploration pages; black-box testing across 14 scenarios achieved a 100% success rate. By turning a large-scale *sanad* KG into a freely explorable front-end, the system addresses the static, desktop-bound limitations of prior work and makes the Islamic scholarly network broadly accessible to researchers, students, and the public.

Future work should compute closeness and betweenness on the full graph using approximation algorithms or parallel computation, incorporate periodic data updates and additional *sanad* sources, add narrator-quality and *sanad*-authenticity dimensions grounded in hadith science, and validate narrator names with expert input to further improve the accuracy of the network and its centrality analysis.

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